LAB 9 : Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES

- To set up a wireless router and build a wireless network environment using Cisco Packet Tracer.
- To connect wireless client devices to the network and distribute IP addresses automatically through DHCP.
- To confirm secure wireless communication by applying WPA2 security settings.

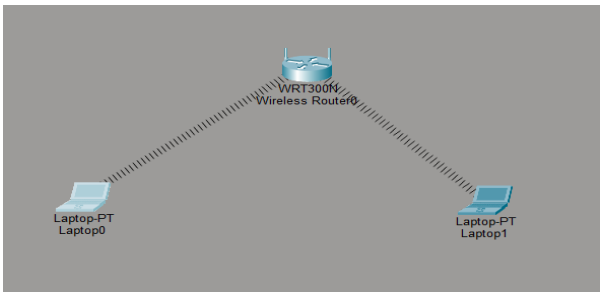
THEORY

A wireless network allows devices to exchange data without the need for physical cables by transmitting signals over radio waves. In this lab, a wireless router was configured to build a wireless local area network (WLAN) that provides internet connectivity, automatic IP address allocation via DHCP, and secured access for wireless clients. The router is responsible for distributing IP addresses to connected devices, which supports smooth and efficient communication across the network.

The network is made identifiable through an SSID, while WPA2 security paired with AES encryption keeps all communications protected from unwanted intrusions. Wireless clients connect to the network by supplying the correct SSID and passphrase, and successful connectivity is then confirmed through network testing.

Packet analysis is used to observe and interpret the flow of data between the router and connected wireless devices. Through Cisco Packet Tracer's simulation and packet inspection features, different packet types — such as DHCP, ARP, and data packets — can be closely examined. This analysis illustrates how devices obtain IP addresses, form connections, and send data over a wireless network. It further assists in confirming successful communication and diagnosing issues, making packet analysis an indispensable part of wireless network implementation.

Network Topology :



Procedure :

Wireless Router Configuration :

Step 1 : Initial Setup

1. Select the Wireless Router device.
2. Open GUI → Setup.
3. Enter the following settings:
   - IP Address: 192.168.0.1
   - Subnet Mask: 255.255.255.0
   - DHCP Server: Enabled

Step 2 : Wireless Network Setup

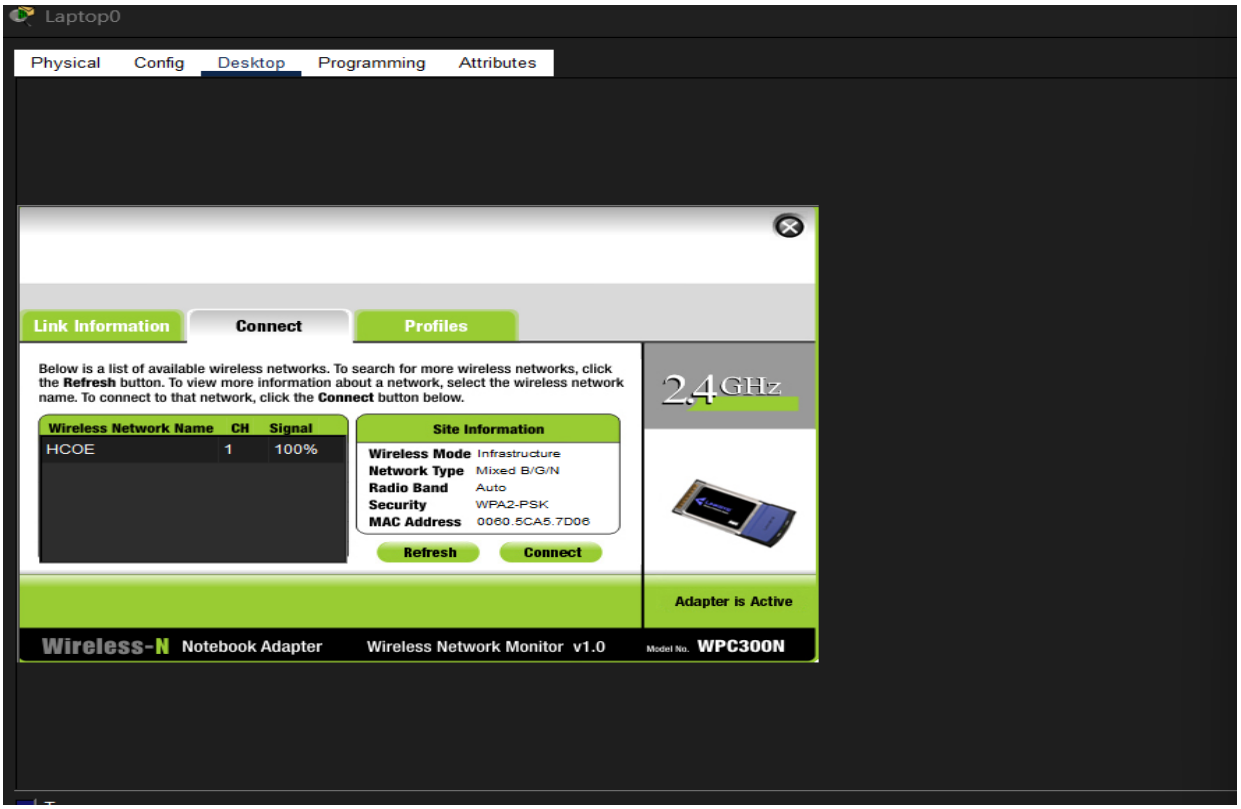
4. Navigate to Wireless → Basic Wireless Settings.
5. Enter the following parameters:
6. Network Mode: Mixed
   - SSID: HCOE
   - Channel: Auto
7. Click Save Settings.

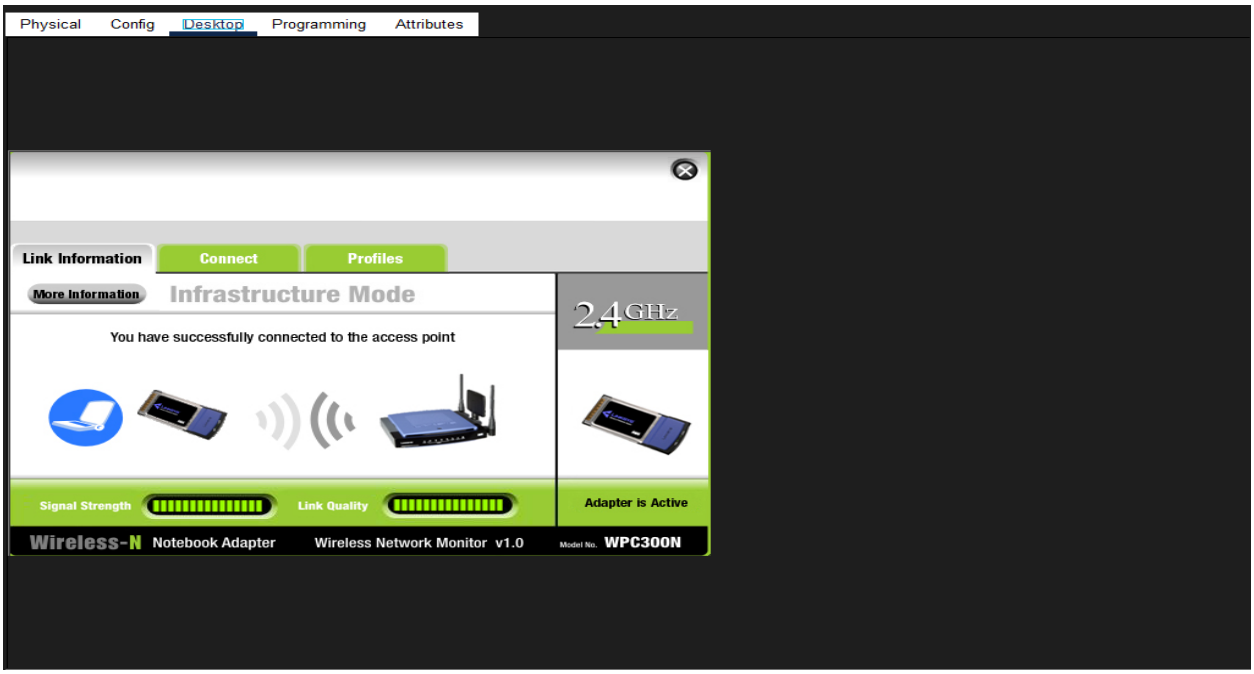
Step 3 : Configuring Wireless Security

8. Navigate to Wireless → Wireless Security.
9. Apply the following settings:
   - Security Mode: WPA2-Personal
   - Encryption: AES
   - Passphrase: hcoe@123

10. Save the configuration

Observation :





RESULT

The configuration successfully created a fully operational wireless network using a WRT300N router. The SSID was defined as "HCOE," and the DHCP server was activated with a gateway address of 192.168.0.1, readying the network to accept client connections. Both Laptop0 and Laptop1 detected the wireless network at full signal strength and connected successfully after authenticating with the WPA2PSK passphrase "hcoe@123." Following authentication, both devices were automatically assigned IP addresses and were able to transmit data packets through the router, verifying that the network was functioning correctly.

DISCUSSION

This lab illustrated how DHCP and WPA2 security work together to create a wireless network that is both functional and secure. Activating the DHCP server allowed the router to automatically distribute IP addresses to both laptops, offering a faster and more dependable setup compared to manual configuration. Applying WPA2-Personal with AES encryption was essential, as it protects the network by enforcing passphrase authentication and blocking unauthorized devices from gaining access.

CONCLUSION

The lab effectively showed that a secure and fully functional wireless network can be built by accurately configuring the SSID, WPA2 security parameters, and DHCP services on the wireless router.